

Algorithms for Data Science

Lecture 3: Algorithms for feature selection (Part 1)



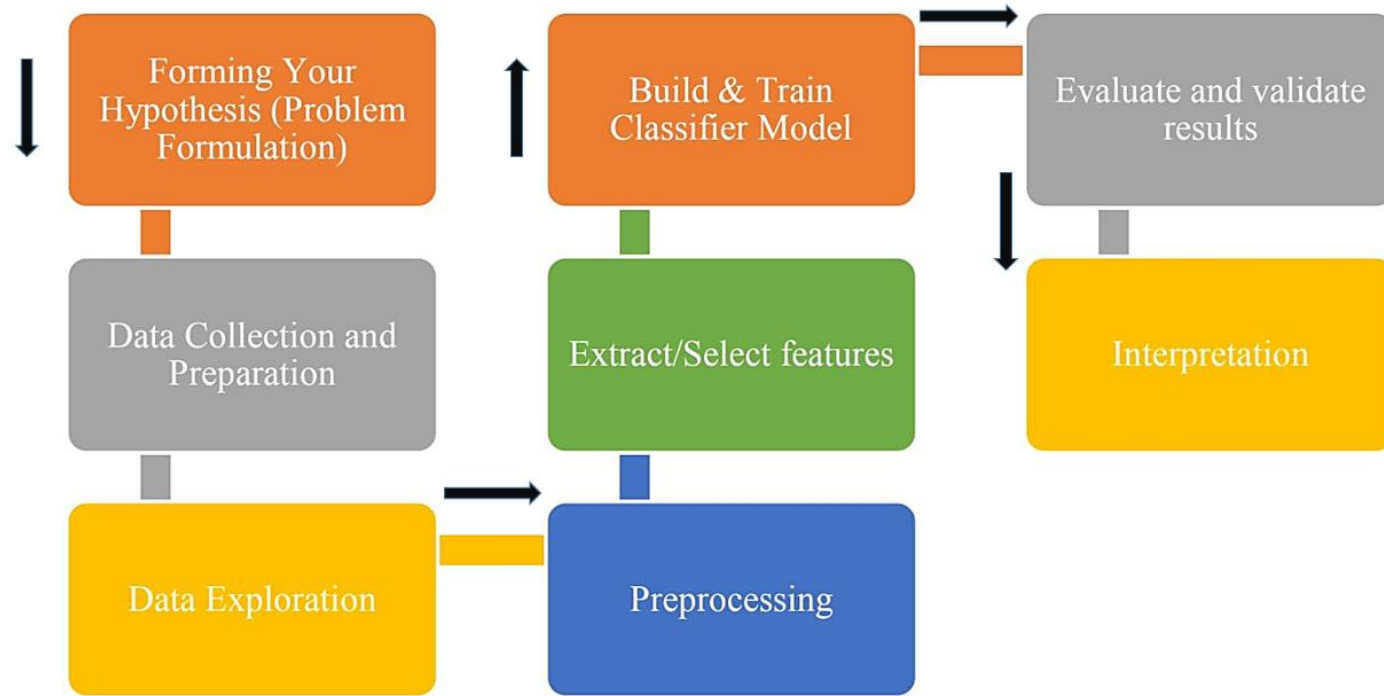


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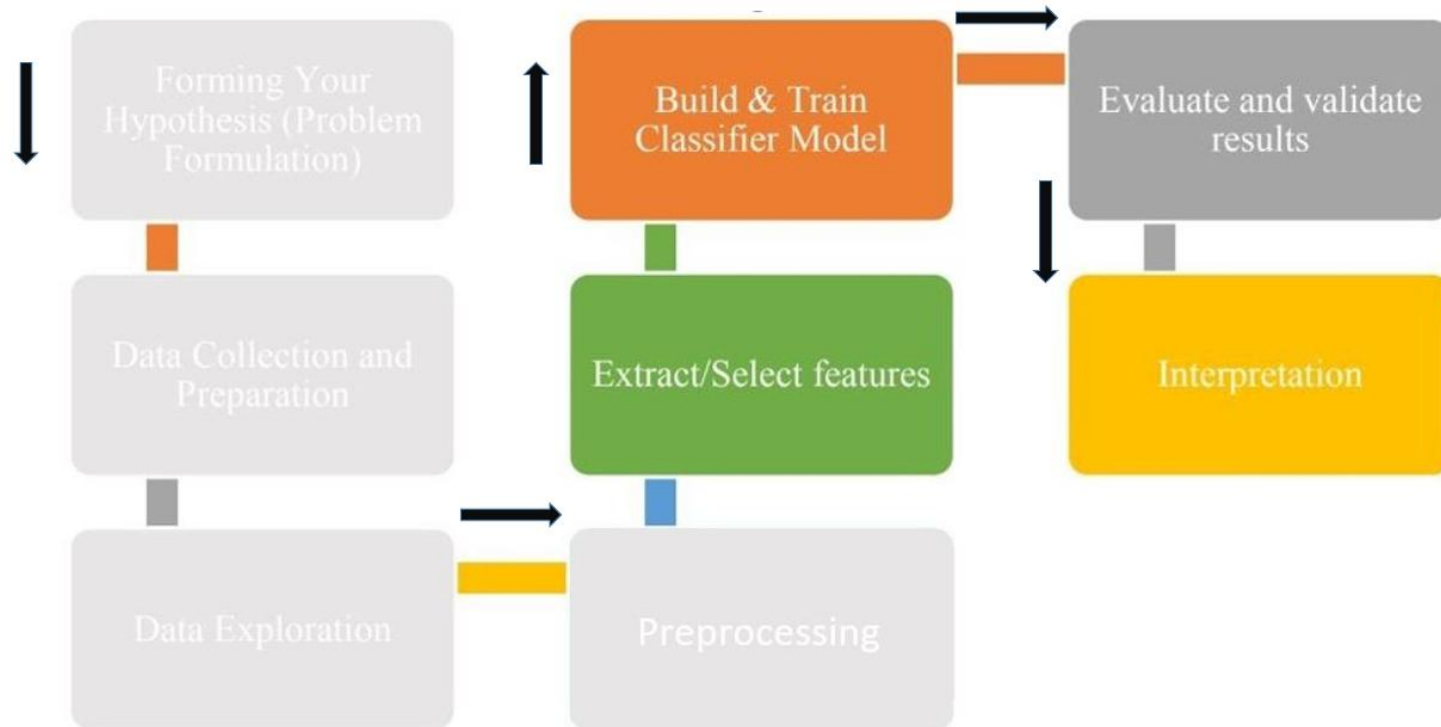
- Feature Reduction
- Feature Selection
- Feature Extraction
- Exploring Feature Selection Approaches



The Data Science Pipeline

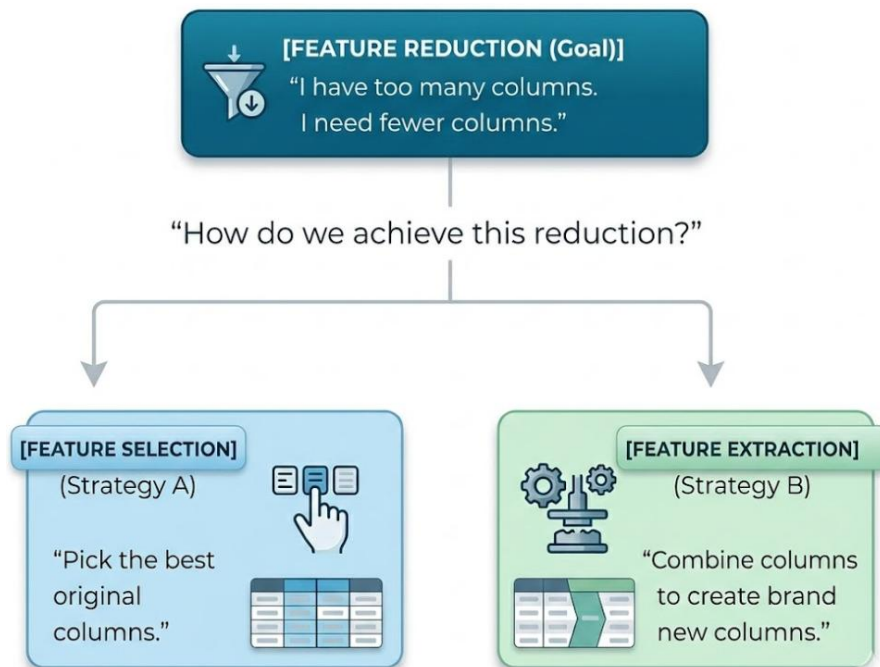


The Data Science Pipeline



Feature Reduction

- Feature Reduction is the process of reducing the number of input variables (features) in a dataset. It is the umbrella term that covers two distinct methods: Feature Selection, and Feature Extraction.



Feature Reduction

- **Why Feature Reduction?**

- ✓ **Reduces Dimensionality:** High-dimensional data can lead to the curse of dimensionality, where the data becomes sparse and models struggle to generalize.
- ✓ **Improves Computational Efficiency:** Fewer features mean faster training and inference times.
- ✓ **Handles Multicollinearity:** Feature reduction can address issues where features are highly correlated.
- ✓ **Visualization:** Reducing features to 2 or 3 dimensions makes it easier to visualize data.



Feature Selection

- **Feature selection:** is the process of selecting a subset of the most relevant features from the original set of features while discarding the irrelevant or less important ones.
- **Common Feature Selection Techniques**
 - Filter Methods
 - Wrapper Methods
 - Embedded Methods



Feature Extraction

- **Feature Extraction:** Mathematically combining multiple columns to create new, fewer columns.
- *Example:* Combining "Height" and "Width" into a single new column called "Area," and then deleting the originals.
- **Common Feature Extraction Techniques**
 - Principal Component Analysis (PCA)
 - Linear Discriminant Analysis (LDA)
 - t-SNE (t-Distributed Stochastic Neighbor Embedding)



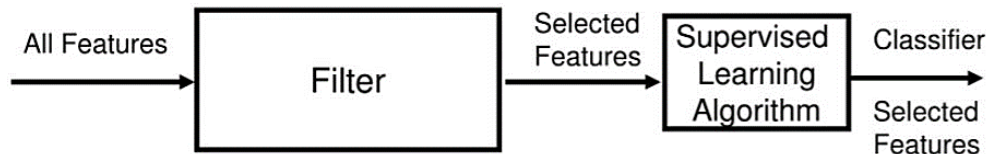
Today the Feature Selection Techniques will be covered



Filter Methods

- **Filter methods** select features before training any model, acting as a pre-processing step that ranks features using statistical tests and retains only the highest-scoring ones.
- These methods work by examining the relationship between each feature and the target variable
- ML Model involvement is none; the ML model is not used at any point during selection. As a result, filter methods are the fastest approach to feature selection.
- **Common Techniques:**
 - ✓ Pearson Correlation
 - ✓ Chi-Square Test

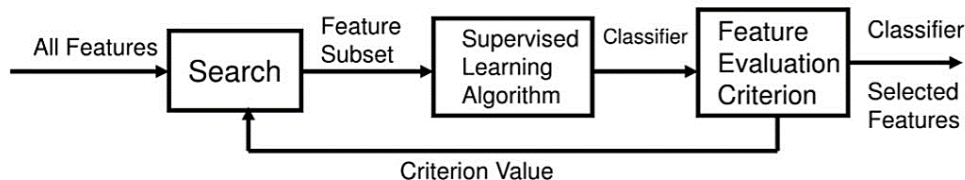
Filter:



Wrapper Methods (1/3)

- **Wrapper methods** select features by training and testing models repeatedly, treating feature selection as a search problem. They experiment with different subsets of features, train a model on each candidate subset, and retain the combination that yields the best performance.
- This approach works by using model accuracy as the feedback score to guide the search for the optimal feature subset. Model involvement is significant, as a specific algorithm—such as a Random Forest—is trained repeatedly throughout the process.
- Consequently, wrapper methods are slow and computationally expensive, but they offer the professional advantage of iterative subset selection directly guided by model performance.

Wrapper:



Wrapper Methods (2/3)

- **Two Types of Wrappers:**

Type	Approach	Algorithms
Deterministic	Follows a fixed, greedy path (add or remove features one by one).	Forward Selection, Backward Elimination, Recursive Feature Elimination (RFE)
Stochastic (Population-Based)	Uses a population of candidate solutions that explore the feature space collectively.	Genetic Algorithms (GA) , Particle Swarm Optimization (PSO)



Wrapper Methods (3/3)

- **Genetic Algorithm (GA):** Mimics natural evolution. It creates a population of feature subsets, evaluates their "fitness" (model accuracy), and breeds new subsets through crossover and mutation until the best subset emerges.
- **Particle Swarm Optimization (PSO):** Mimics social behavior like bird flocking or fish schooling. Each "particle" represents a feature subset and moves through the search space, adjusting its position based on its own best-known position and the swarm's best-known position. Over time, the swarm converges on the optimal feature subset.



Embedded Methods

- **Embedded methods** select features during model training, with feature selection built directly into the learning process itself.
- The algorithm decides which features are important while it learns, containing internal mechanisms, such as regularization penalties or importance scores—that penalize or ignore useless features during training.
- Model involvement is intrinsic, as selection happens automatically within a single training session. This makes embedded methods fast, requiring only one model training cycle. From a professional standpoint, they offer intrinsic feature selection integrated into the model learning phase.
- Common algorithms include Random Forest Feature Importance, and Gradient Boosting (XGBoost, LightGBM) Importance.



Lets start with one of the filter methods:

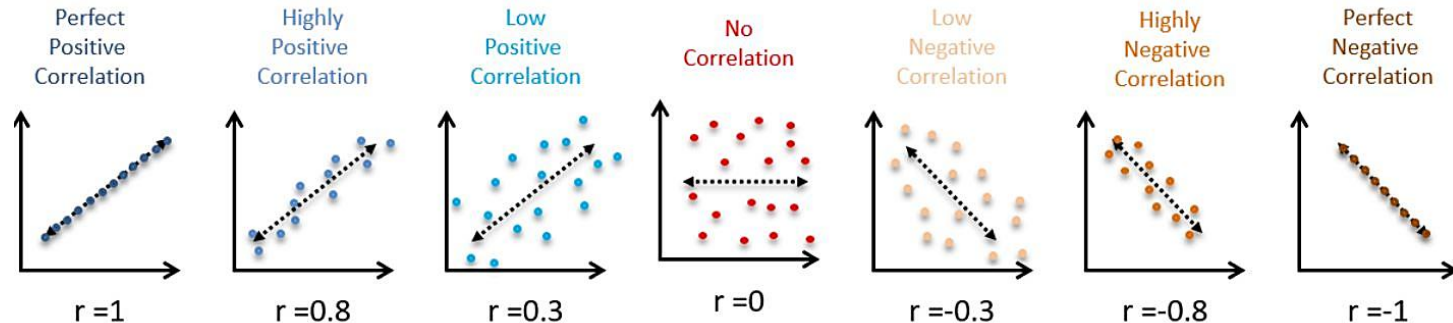
Pearson Correlation



Important Terms- Covariance (1/2)

- The covariance is a measure of how two variables change together. It indicates the direction of the linear relationship between the variables:
 - ✓ If the covariance is positive, the variables tend to increase or decrease together.
 - ✓ If the covariance is negative, one variable tends to increase when the other decreases.
 - ✓ If the covariance is zero, there is no linear relationship between the variables.

Scatter Plots & Correlation Examples



Important Terms- Covariance (2/2)

Mathematical Definition

For two random variables X and Y , the **covariance** is defined as:

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_X)(y_i - \mu_Y)$$

where:

- μ_X = mean of X ,
- μ_Y = mean of Y ,



Filter Methods- Pearson Correlation

Pearson correlation measures the linear relationship between a feature and the target variable. The correlation coefficient (r) ranges from -1 to 1 , where:

- $r = 1$: Perfect positive linear relationship.
- $r = -1$: Perfect negative linear relationship.
- $r = 0$: No linear relationship.

Formula:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

where:

- $\text{Cov}(X, Y)$ is the covariance between X and Y .
- σ_X and σ_Y are the standard deviations of X and Y , respectively.



Filter Methods- Pearson Correlation

Steps:

1. Calculate the mean of each feature and the target variable.
2. Compute the covariance and standard deviations.
3. Calculate the correlation coefficient for each feature.

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_X)(y_i - \mu_Y)$$

where:

- μ_X = mean of X ,
- μ_Y = mean of Y ,

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$



Filter Methods- Pearson Correlation (example 1)

Consider the following dataset with **3 features** (F_1, F_2, F_3) and a **target variable** (Y):

F_1	F_2	F_3	Y
1	5	10	2
2	6	15	4
3	7	20	5
4	8	25	4
5	9	30	6



Filter Methods- Pearson Correlation (example 1)

First, compute the mean of each feature (F_1, F_2, F_3) and the target variable (Y):

$$\bar{F}_1 = \frac{1 + 2 + 3 + 4 + 5}{5} = 3$$

$$\bar{F}_2 = \frac{5 + 6 + 7 + 8 + 9}{5} = 7$$

$$\bar{F}_3 = \frac{10 + 15 + 20 + 25 + 30}{5} = 20$$

$$\bar{Y} = \frac{2 + 4 + 5 + 4 + 6}{5} = 4.2$$



Filter Methods- Pearson Correlation (example 1)

For each feature and the target variable, calculate the deviations from the mean:

F_1	F_2	F_3	Y	$F_1 - \bar{F}_1$	$F_2 - \bar{F}_2$	$F_3 - \bar{F}_3$	$Y - \bar{Y}$
1	5	10	2	$1 - 3 = -2$	$5 - 7 = -2$	$10 - 20 = -10$	$2 - 4.2 = -2.2$
2	6	15	4	$2 - 3 = -1$	$6 - 7 = -1$	$15 - 20 = -5$	$4 - 4.2 = -0.2$
3	7	20	5	$3 - 3 = 0$	$7 - 7 = 0$	$20 - 20 = 0$	$5 - 4.2 = 0.8$
4	8	25	4	$4 - 3 = 1$	$8 - 7 = 1$	$25 - 20 = 5$	$4 - 4.2 = -0.2$
5	9	30	6	$5 - 3 = 2$	$9 - 7 = 2$	$30 - 20 = 10$	$6 - 4.2 = 1.8$



Filter Methods- Pearson Correlation (example 1)

For F_1 :

$$\begin{aligned}\text{Cov}(F_1, Y) &= \frac{(-2)(-2.2) + (-1)(-0.2) + (0)(0.8) + (1)(-0.2) + (2)(1.8)}{5} \\ &= \frac{4.4 + 0.2 + 0 - 0.2 + 3.6}{5} = \frac{8.0}{5} = 1.6\end{aligned}$$

For F_2 :

$$\begin{aligned}\text{Cov}(F_2, Y) &= \frac{(-2)(-2.2) + (-1)(-0.2) + (0)(0.8) + (1)(-0.2) + (2)(1.8)}{5} \\ &= \frac{4.4 + 0.2 + 0 - 0.2 + 3.6}{5} = \frac{8.0}{5} = 1.6\end{aligned}$$

For F_3 :

$$\begin{aligned}\text{Cov}(F_3, Y) &= \frac{(-10)(-2.2) + (-5)(-0.2) + (0)(0.8) + (5)(-0.2) + (10)(1.8)}{5} \\ &= \frac{22 + 1 + 0 - 1 + 18}{5} = \frac{40.0}{5} = 8.0\end{aligned}$$



Filter Methods- Pearson Correlation (example 1)

The standard deviation of a feature (F_i) or the target (Y) is

For F_1 :

$$\sigma_{F_1} = \sqrt{\frac{(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}{5}} = \sqrt{\frac{4 + 1 + 0 + 1 + 4}{5}} = \sqrt{\frac{10}{5}} = \sqrt{2} \approx 1.414$$

For F_2 :

$$\sigma_{F_2} = \sqrt{\frac{(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}{5}} = \sqrt{\frac{4 + 1 + 0 + 1 + 4}{5}} = \sqrt{\frac{10}{5}} = \sqrt{2} \approx 1.414$$

For F_3 :

$$\sigma_{F_3} = \sqrt{\frac{(-10)^2 + (-5)^2 + 0^2 + 5^2 + 10^2}{5}} = \sqrt{\frac{100 + 25 + 0 + 25 + 100}{5}} = \sqrt{\frac{250}{5}} = \sqrt{50}$$

For Y :

$$\sigma_Y = \sqrt{\frac{(-2.2)^2 + (-0.2)^2 + 0.8^2 + (-0.2)^2 + 1.8^2}{5}} = \sqrt{\frac{4.84 + 0.04 + 0.64 + 0.04 + 3.24}{5}} = \sqrt{\frac{8.8}{5}} = \sqrt{1.76} \approx 1.327$$



Filter Methods- Pearson Correlation (example 1)

Compute Pearson Correlation

Using the formula:

$$r = \frac{\text{Cov}(F_i, Y)}{\sigma_{F_i} \cdot \sigma_Y}$$

For F_1 :

$$r_{F_1, Y} = \frac{1.6}{1.414 \cdot 1.327} \approx \frac{1.6}{1.876} \approx 0.853$$

For F_2 :

$$r_{F_2, Y} = \frac{1.6}{1.414 \cdot 1.327} \approx \frac{1.6}{1.876} \approx 0.853$$

For F_3 :

$$r_{F_3, Y} = \frac{8.0}{7.071 \cdot 1.327} \approx \frac{8.0}{9.38} \approx 0.853$$



Filter Methods- Pearson Correlation (example 1)

- The Pearson Correlation coefficients for all features are approximately 0.853, indicating a **strong positive linear relationship** with the target variable Y .
- If we set a threshold (e.g., $|r| > 0.5$), all features (F_1, F_2, F_3) would be selected.



Filter Methods- Pearson Correlation (example 2)

Suppose we have the following dataset with **4 features** (F_1, F_2, F_3, F_4) and a **target variable** (Y):

F_1	F_2	F_3	F_4	Y
1	5	10	0	1
2	6	15	1	0
3	7	20	0	1
4	8	25	1	0

Results:

- $r_{F_1,Y} = 0$
- $r_{F_2,Y} = -1$ (perfect negative correlation)
- $r_{F_3,Y} = 0$
- $r_{F_4,Y} = 1$ (perfect positive correlation)

Feature Selection:

- Select features with the highest absolute correlation values: F_2 and F_4 .



Lets start with one of the wrapper methods:

Genetic Algorithm



Wrapper Methods- Genetic Algorithm

Genetic Algorithm (GA) is a population-based optimization algorithm inspired by the process of natural selection and evolution which means those species who can adapt to changes in their environment are able to survive and reproduce and go to next generation. In simple words, they simulate “survival of the fittest” among individual of consecutive generation for solving a problem.

The **Genetic Algorithm (GA)** is a popular optimization technique that can be combined with wrapper methods to search for the best subset of features.

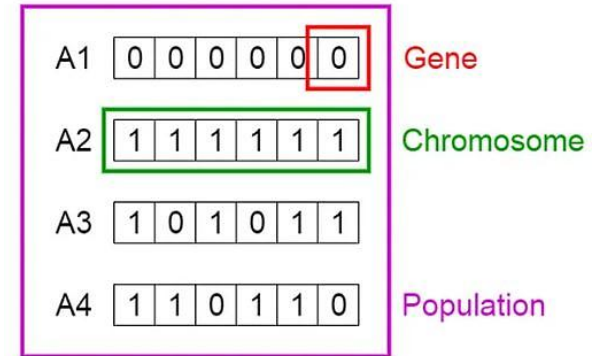


Genetic Algorithm

- Each generation consist of a population of individuals, and each individual represents a point in search space and possible solution. Each individual is represented as a string of character/integer/float/bits. This string is analogous to the Chromosome.

- **Search space:**

The population of individuals are maintained within search space. Each individual represents a solution in search space for given problem. Each individual is coded as a finite length vector (analogous to chromosome) of components. These variable components are analogous to Genes. Thus, a chromosome (individual) is composed of several genes (variable components).



Genetic Algorithm

- Phases are considered in a genetic algorithm.
 - ✓ Problem encoding (gene, chromosome)
 - ✓ Initial population
 - ✓ Fitness score
 - ✓ Selection operator
 - ✓ Crossover operator
 - ✓ Mutation operator
 - ✓ Replacement
 - ✓ Termination



Genetic Algorithm

- **Problem encoding:** When encoding a problem for a genetic algorithm (GA), the goal is to represent potential solutions (individual) in a way that can be manipulated by genetic operators like crossover and mutation. The encoding scheme is crucial because it directly impacts the performance and effectiveness of the GA.
- **Some common individual's encodings:**
 - ✓ Binary encoding (0101...1100)
 - ✓ Real-Valued Encoding [3.2, -1.5]
 - ✓ Lists of rules $R_1 R_2 R_3 R_{22} R_{23}$



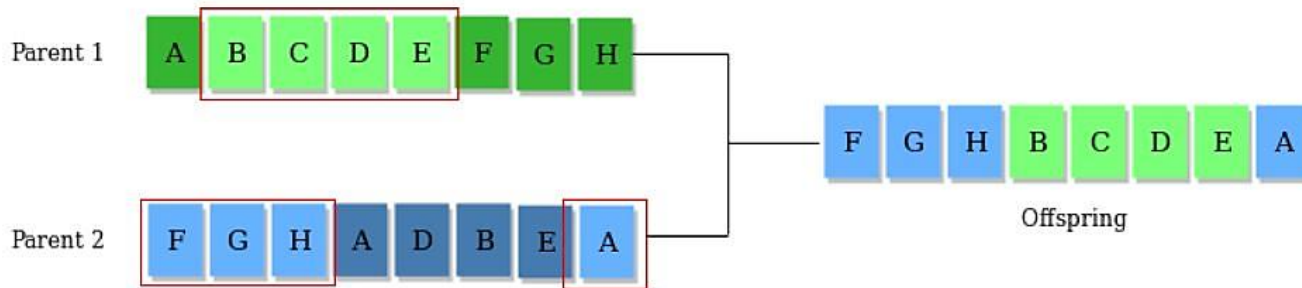
Genetic Algorithm

- **Initial population:** Generate an initial population of random individuals.
- **Fitness score:** Each individual (solution) in the population is assigned a fitness score, which measures how good it is at solving the problem. Individuals with higher fitness scores are more likely to be selected for reproduction.
- **Selection:** The idea is to give preference to the individuals with good fitness scores and allow them to pass their genes to successive generations. There are many selection ways but the simplest way is to rank all the individuals from best to worst and choose the best one.



Genetic Algorithm

- **Crossover:** Combine two individuals to produce offspring. Two individuals are selected using selection operator and crossover sites are chosen randomly. Then the genes at these crossover sites are exchanged thus creating a completely new individual (offspring).
- For example:



Genetic Algorithm

- **Mutation:** The key idea is to insert random genes in offspring to maintain the diversity in the population to avoid premature convergence.

Before Mutation



- For example:

After Mutation



- **It is not applied to all offspring.** The simplest way to choose individuals for mutation in a genetic algorithm is to randomly **select individuals from the population with a fixed probability.**



Genetic Algorithm

• Mutation probability example

Suppose you have a population of 10 individuals and a mutation probability $p_m = 0.1$ (10%). Here's how you might select individuals for mutation:

- For each individual in the population:
 - Generate a random number r uniformly distributed between 0 and 1.
 - Compare r with p_m :
 - If $r \leq p_m$, mutate the individual.
 - If $r > p_m$, leave the individual unchanged.

```
Individual 1: r = 0.03 (mutate)
Individual 2: r = 0.15 (skip)
Individual 3: r = 0.08 (mutate)
Individual 4: r = 0.45 (skip)
Individual 5: r = 0.92 (skip)
Individual 6: r = 0.11 (skip)
Individual 7: r = 0.01 (mutate)
Individual 8: r = 0.67 (skip)
Individual 9: r = 0.09 (mutate)
Individual 10: r = 0.22 (skip)
```



Genetic Algorithm

- **Replacement:** The new candidate solutions replace the least fit individuals in the population.
- **Termination:** The algorithm terminates when a stopping criterion is met, such as a maximum number of generations or a satisfactory level of fitness



Genetic Algorithm

- Produce an initial population of individuals
- Evaluate the fitness of all individuals
- While termination condition not met do
 - select fitter individuals for reproduction
 - recombine between individuals
 - mutate individuals
 - evaluate the fitness of the modified individuals
 - generate a new population
- End while



Genetic Algorithm- Example

Problem Setup:

- Suppose we have a dataset with 5 features: F_1, F_2, F_3, F_4, F_5 .
- We want to select the best subset of 3 features that maximizes the accuracy of a classifier.



Genetic Algorithm- Example

Genetic Algorithm Steps:

1. Initialization:

- Create an initial population of chromosomes, where each chromosome represents a possible subset of features.
- Each chromosome is a binary string of length 5, where **1** indicates the feature is selected and **0** indicates it is not.
- Example population:
 - Chromosome 1: **1 0 1 0 1** (Features F_1, F_3, F_5)
 - Chromosome 2: **0 1 1 1 0** (Features F_2, F_3, F_4)
 - Chromosome 3: **1 1 0 0 1** (Features F_1, F_2, F_5)
 - Chromosome 4: **0 0 1 1 1** (Features F_3, F_4, F_5)



Genetic Algorithm- Example

2. **Fitness Evaluation:**

- Evaluate the fitness of each chromosome by training a classifier using the selected features and measuring the accuracy.
- Suppose the accuracies are:
 - Chromosome 1: 85%
 - Chromosome 2: 80%
 - Chromosome 3: 90%
 - Chromosome 4: 75%



Genetic Algorithm- Example

3. Selection:

- Select the **two fittest chromosomes**:
 - Chromosome 3: 1 1 0 0 1 (90%)
 - Chromosome 1: 1 0 1 0 1 (85%)



Genetic Algorithm- Example

4. Crossover:

- Perform crossover on the two selected chromosomes to create offspring.
- Use **single-point crossover** (randomly choose a crossover point):
 - Parent 1: 1 1 0 0 1 (Chromosome 3)
 - Parent 2: 1 0 1 0 1 (Chromosome 1)
 - Crossover point: After the 2nd bit
 - Offspring 1: 1 1 | 1 0 1 → 1 1 1 0 1 (Features F_1, F_2, F_3, F_5)
 - Offspring 2: 1 0 | 0 0 1 → 1 0 0 0 1 (Features F_1, F_5)



Genetic Algorithm- Example

5. Mutation:

- Apply mutation to **only half of the offspring** (e.g., randomly select one offspring to mutate):
 - Randomly select Offspring 1 for mutation.
 - Flip one bit in Offspring 1 (e.g., flip the 4th bit):
 - Offspring 1: $1\ 1\ 1\ 0\ 1 \rightarrow 1\ 1\ 1\ 1\ 1$ (Features F_1, F_2, F_3, F_4, F_5)
 - Offspring 2 remains unchanged: $1\ 0\ 0\ 0\ 1$ (Features F_1, F_5)



Genetic Algorithm- Example

6. Replacement:

- Replace the old population with the new population (offspring and possibly some elites):
 - Chromosome 1: 1 1 1 1 1 (Features F_1, F_2, F_3, F_4, F_5)
 - Chromosome 2: 1 0 0 0 1 (Features F_1, F_5)
 - Chromosome 3: 1 1 0 0 1 (Features F_1, F_2, F_5) (retained as one of the fittest)
 - Chromosome 4: 1 0 1 0 1 (Features F_1, F_3, F_5) (retained as one of the fittest)

7. Repeat:

- Repeat the process for multiple generations until convergence or a stopping criterion is met.



Any Question?

